

Haematopoiesis in the era of advanced single-cell technologies

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The molecular and functional characterization of single cells at scale has emerged as a key driver to unravel tissue biology. Thus, it is important to understand the strengths and limitations of transcriptomic approaches, molecular barcoding and functional assays used to study cellular properties at the single-cell level. Here, we review recent relevant work from the haematopoietic system and discuss how to interpret and integrate data obtained with different technologies.

The haematopoietic system is one of the most thoroughly characterized differentiation hierarchies in mammals, and haematopoiesis has long served as a paradigm for how stem cells maintain and regenerate complex, self-renewing tissues. For decades, the study of haematopoiesis has been facilitated by analysis at the single-cell level, including *in vitro* and *in vivo* clonal assays for haematopoietic stem/progenitor cell function, and through multi-parameter flow cytometry and cell sorting, together allowing the systematic characterization of highly purified cell populations^{1,2}. Using these single-cell methodologies, haematopoietic stem cells (HSCs), defined as cells capable of reconstituting multi-lineage haematopoiesis in lethally irradiated recipient mice upon transplantation, were first identified by McCulloch and Till³, isolated by Spangrude et al.⁴ and subsequently purified to increasing homogeneity^{5,6}. Similarly, the discovery of haematopoietic cytokines⁷ enabled the development of lineage-specific clonal colony-forming cell assays, allowing progenitor cells to be isolated and their lineage potential assessed at the single-cell level. This powerful combination of clonal and single-cell technologies generated hierarchical models of the haematopoietic system^{8–10}, underpinning numerous molecular and genetic studies into the development and regulation of haematopoiesis and the aetiology of haematological disorders¹. In all proposed hierarchies, haematopoiesis originates in multipotent repopulating HSCs, and lineage development occurs via discrete and increasingly lineage-restricted progenitor populations. However, as predominantly purified cell populations were analysed as opposed to individual cells, functional heterogeneity within the identified populations might have been underestimated. In addition, although transplantation of purified prospective HSC populations into irradiated recipient mice is a powerful assay to evaluate repopulating HSCs at the single-cell level, it may not be able to de-convolute HSC heterogeneity and function in steady-state haematopoiesis.

Recently, several studies have explored emerging single-cell technologies that have the potential to overcome many of these limitations^{2,11–15}. Single-cell whole-transcriptome analysis has allowed molecular heterogeneity to be systematically explored across the haematopoietic system². In addition, *in vivo* barcoding has emerged as a means to study the fate and contribution of individual stem and progenitor cells within larger populations, including their roles in unperturbed haematopoiesis^{11–13,15}. Together, these innovative single-cell technologies promise to considerably advance our understanding of haematopoiesis. However, as for the more classical single-cell technologies, recent single-cell approaches have

technical and conceptual limits that need to be considered when interpreting the results they generate. To this end, we discuss the key concepts used to functionally define haematopoietic stem and progenitor cells and how they apply to data generated by both classical and emerging technologies.

Current technologies and their limitations

Defining cellular pathways and populations through which haematopoietic lineages are generated remains an important task; lineage relationships provide important information about the evolutionary ontogeny of haematopoietic cell types. Knowledge of lineage branch points facilitates the unravelling of molecular mechanisms that specify cell fates. Furthermore, progenitor cells are templates for cellular transformation in many haematopoietic malignancies¹⁶. Elucidation of their molecular characteristics allows accurate comparison of normal versus transformed progenitors and the identification of transformation-specific therapeutic targets. Particular challenges exist when studying progenitor behaviour. Many progenitor cells are by nature short lived with little or no self-renewal potential and limited repopulation ability in a transplantation setting, in particular, for short-lived, non-lymphoid cell types¹⁷. As a result, progenitor populations have generally been characterized using *in vitro* colony-forming cell assays, other clonal assays and transient *in vivo* repopulation.

These classical technologies led to the identification of multipotent cell populations restricted to lymphoid lineage potential (common lymphoid progenitor (CLP))⁹ or myeloerythroid lineage potential (common myeloid progenitor (CMP))⁸. Data sets indicated a hierarchical model in which the initial lineage bifurcation downstream of the HSC would strictly separate the lymphoid and myeloerythroid lineages, with subsequent sequential branching leading to specification of individual lineages via intermediate granulocyte-macrophage progenitors (GMPs) and megakaryocyte-erythroid progenitors (Fig. 1a). Subsequent studies identified additional subpopulations with predominantly erythroid (colony-forming unit-erythroid (CFU-E) and preCFU-E) or myeloid (pre-granulocyte-macrophage progenitor (preGM)) potential¹⁰, progenitors with combined lymphoid and myeloid potential that lack erythroid/megakaryocyte potential (lymphoid-primed multipotent progenitor (LMPP))^{18,19} and amended definitions of the CMP^{20,21}, resulting in a revised hierarchical model (Fig. 1b). However, further refinement of haematopoietic differentiation pathways will be required, in particular, to account for the possibility of cell types arising through

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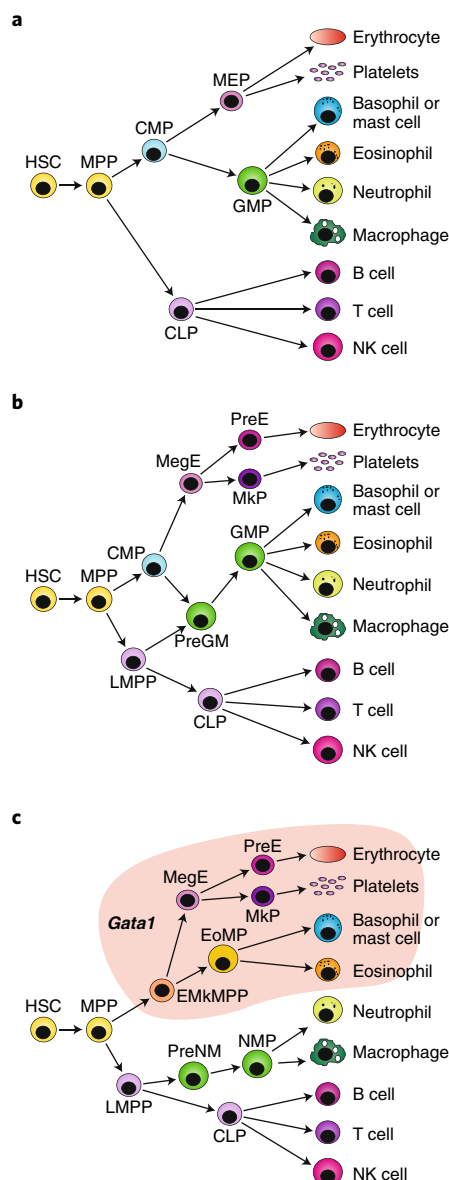


Fig. 1 | Alternative hierarchical models of haematopoiesis. a, CMP-CLP model, based on the prospective isolation of cell populations exclusively committed to myeloid/erythroid (CMP) or lymphoid (CLP) fates. **b**, CMP-LMPP model, incorporating the identification of a myeloid/LMPP, as well as the further fractionation of committed myeloid and megakaryocyte-erythroid progenitors (MEPs). **c**, Erythroid-megakaryocyte-primed multipotent progenitor (EMkMPP)-LMPP model, in which myelopoiesis proceeds via distinct *Gata1*-positive (generating eosinophils, mast cells or basophils) and *Gata1*-negative pathways (generating neutrophils or monocytes). The shaded area indicates the *Gata1* expression domain. MkP, megakaryocyte progenitor; MegE, pre-megakaryocyte-erythroid progenitor; preE, pre-colony-forming erythroid progenitor; EoMP, eosinophil-mast cell progenitor; NMP, neutrophil-monocyte progenitor; preNM, pre-neutrophil-monocyte progenitor; NK, natural killer.

more than one progenitor route, as exemplified by the specification of distinct monocyte populations from GMPs and monocyte-dendritic progenitors²².

For decades, stem and progenitor cells have been annotated based on their behaviour in the traditional haematopoietic single-cell assays described above. However, we now need to examine how these annotations should be (re-)defined in light of emerging

single-cell technologies (Table 1). An important consideration when interpreting experimental data is how properties of the assays may influence the readout and therefore limit the conclusions that can be reached. For example, HSC transplantation assays evaluate the ability of putative HSCs to repopulate the haematopoietic system, but do not address the extent to which the transplanted cells contribute to haematopoiesis in their native, steady-state setting^{12,13,15}. Indeed, highly quiescent HSCs (which by inference generate little or no cellular output) have the highest repopulating capacity upon transplantation^{23,24}. Likewise, *in vivo* and *in vitro* lineage potential of the same single HSC may differ greatly²⁵, as may its lineage fate when using different *in vivo* settings.

For progenitors, a key issue is the measurement of lineage potential and lineage fate and establishing the relationship between the two. In this case, the limited proliferative potential and distinct, transient kinetics in lineage replenishment make the *in vivo* readout of multiple lineage fates particularly challenging, and *in vitro* colony-forming cell and other clonal assays remain the preferred tool¹⁷. However, individual progenitor cells do not necessarily adopt all possible lineage outcomes *in vitro*²⁶. Thus, although a positive lineage readout for an individual progenitor demonstrates its ability to acquire this particular fate, a negative readout does not necessarily demonstrate the absence of that potential. Consequently, lineage restriction can only be demonstrated through the consistent absence of specific lineage readouts across an entire cell population, requiring its prospective purification. Importantly, similar to what is observed for lineage-biased HSCs²⁵, the detected fate of a progenitor might not adequately represent its full lineage potential, as not all fate options may be utilized under a given condition. Thus, establishing the definite potency of a progenitor at the single-cell level is a considerable challenge.

Fate mapping through single-cell transplantation

HSC heterogeneity was initially identified by careful tracking of myeloid and lymphoid lineage output from transplanted single HSC-derived cell populations or single HSCs^{27,28}. These studies observed HSC clones that generated a higher proportion of myeloid cells (M biased) or lymphoid cells (L biased) than the HSC population average, a bias stable during up to three serial transplantations^{27,28}. Thus, they identified HSC lineage bias as an intrinsically programmed HSC property. Although transplantation of single cells with a CD45.2 allotype into congenic CD45.1 recipient mice allowed specific tracking of CD45.2 myeloid (M; granulocyte-monocyte), B and T lymphoid cells, mature erythrocytes (E) and platelets (P) lack CD45 expression and their replenishment could therefore not be tracked in these previous studies^{27,28}. Additional work using a transgenic reporter to track P-lineage replenishment by transplanted HSCs suggested that a large fraction of long-term HSCs (LT-HSCs; previously classified as M biased) are rather P or PM biased²⁹ and that functional HSC P bias increases significantly with age³⁰. Other studies, using a fluorescent reporter to detect all lineage output (P, E, M, B and T), identified progenitors capable of extended, lineage-restricted P, PE and PEM repopulation³¹. However, most of these progenitors did not sustain stable lineage output in secondary recipients, distinguishing them from LT-HSCs with extensive self-renewal³¹. Although their lineage potential was not investigated *in vitro*, it was concluded that these were lineage-committed oligopotent and unipotent progenitors with sustained self-renewal potential, and therefore that stem cell properties of self-renewal and multipotency could be segregated³¹. More recent single HSC transplantation studies identified LT-HSCs fully and stably restricted to P, PE, PEM or PEMB fates (but not to any other lineage combinations)²⁵. The previously identified, M-biased HSCs (based on tracking of myeloid and lymphoid, but not platelet and erythroid, output) contributed more extensively to replenishment of platelets than to myeloid reconstitution. Although stably fate-restricted *in vivo* HSCs

Table 1 | Key concepts and their use in single-cell biology

Concept	Definition	Assay (or assays)	Limitations	Recent single-cell-technology-based insights	Ref.
Repopulating stem cell	A cell that upon transplantation can stably repopulate all or part of the haematopoietic lineages	Transplantation of prospective HSCs into a suitably conditioned recipient and measurement of cellular output	The detection of HSCs may be limited by the technical ability to detect all lineage outputs. The behaviour of HSCs in transplantation assays may not reflect their fate in their original unperturbed environment	Repopulating HSCs is heterogeneous, both in terms of lineage bias and self-renewal capacity	24,25,31
Self-renewal	The ability of a cell to perpetuate itself during extended cell divisions	Serial transplantation or long-term culture of putative self-renewing cells	Serial transplantation of non-purified HSCs (for example, total bone marrow) can propagate long-lived lymphoid cells in the absence of detectable HSCs, and transplantation may not detect all cells that are capable of self-renewal	Genetic barcoding and Cre-fate mapping studies are compatible with non-repopulating cells that also possess substantial self-renewal in unperturbed adult haematopoiesis	12,35
Lineage potential and/or lineage commitment	The capacity of a cell to differentiate along a certain lineage (or lineages) under appropriate conditions. Lineage commitment is loss of lineage potential that restricts a cell to a subset of its original potential	In vitro clonal or colony-forming or in vivo transplantation assays that measure the relevant potential	Not all lineage potential will necessarily be detected. Thus, a negative finding for a particular cell does not demonstrate that the potential was not present. Confirmation of loss of a potential therefore requires the prospective purification of a cell population and consistent negative readout	Refinement of progenitor purification and improved in vitro and in vivo assays and readout methods have allowed more accurate understanding of how lineages segregate	10,18
Lineage fate	The cellular fate (or fates) that a stem cell or progenitor cell will give rise to	Lineage fate is specific to a particular condition. It can only be measured under these (typically in vivo) conditions and lineage fate may change if the conditions are altered	Lineage fate cannot be inferred from assays that remove cells from the relevant environment. As such, fate during unperturbed haematopoiesis cannot be measured in vitro or by re-transplantation	The lineage fate of HSCs may be distinct from their lineage potential (through lineage bias); progenitor fate can be dependent on the assay conditions	25,26
Lineage bias	A term applied to HSCs that display a stable restriction or strong bias of their lineage output towards particular cellular fates	Lineage bias of multipotent HSCs has been demonstrated in vivo in serial transplantations initiated by a single HSC	In vivo lineage bias cannot be inferred from in vitro measurement of lineage potential, as biased or restricted HSCs can retain in vitro potential that does not apply in vivo. In vivo lineage bias may be context dependent	Improved technology to measure HSC outputs, both in transplantation assays and in unperturbed haematopoiesis, has expanded the known repertoire of biased HSCs and, in particular, identified platelet-biased or platelet-restricted HSCs	13,25
Lineage priming	The expression by a multipotent or oligopotent stem/progenitor cell of genes associated with the terminal differentiation of a specific lineage (or lineages)	Single-cell gene expression analysis	Cells may have the potential for different lineages that is not measurable by gene expression analysis, for example, lymphoid potential of HSCs	Deep scRNAseq has identified new putative lineage-restricted progenitors, as well as uncommitted cells with mixed lineage states	44,45

The definitions of key concepts relating to lineage fate and potential are listed, along with the assays used to define them and the key issues that limit data interpretation. Important single-cell-based studies using these technologies are indicated.

remained multipotent in all cases, as lineage potential that was not apparent during in vivo transplantation was revealed upon in vitro culture and, in some instances, also upon secondary transplantation²⁵. Thus, multipotency seems a general property of LT-HSCs, and HSC in vivo fate restriction is guided by strict rules that determine possible and hierarchically related lineage fate combinations.

Studying native haematopoiesis through in vivo fate mapping

Single-cell transplantation has proven highly valuable in the unravelling of HSC heterogeneity. However, previous work also indicates

that full elucidation of HSC properties will require a combination of assays, including both in vivo and in vitro single-cell-based analysis. Even though transplantation assays can accurately identify repopulating HSCs, they do not necessarily inform on their fate during native haematopoiesis. Some HSCs may be highly quiescent and therefore contribute minimally to steady-state haematopoiesis while demonstrating high repopulating activity upon transplantation^{23,24}. An alternative method, viral genetic barcoding, has previously been used to track single HSC clones following transplantation¹¹. More recently, to measure steady-state haematopoietic output at

single-cell resolution, cells have been barcoded in vivo using Sleeping Beauty-mediated transposition to generate unique insertions that can be identified through ligation-mediated PCR^{12,13}. Barcoding is a powerful tool to assign clonal relationships between mature haematopoietic cells and stem/progenitor cells. These studies detected clones originating from HSCs that seemed restricted to platelet lineage output¹³, consistent with the P-biased multipotent HSCs identified by single-cell transplantation^{25,29}, and that these are active during steady-state haematopoiesis. Interestingly, Sawai et al.³² observed clearly distinct kinetics of lineage repopulation from lineage-traced HSCs, with platelets being the most rapidly reconstituted lineage. As these findings correlate well with the observation that HSC barcoding initially labels mostly platelet lineage progenitors, they also raise the possibility that pervasive P-biased HSC readouts at early time points after barcoding¹³ could reflect the more rapid replenishment kinetics of this lineage. Indeed, most barcoded HSCs with early P-bias displayed multi-lineage repopulation upon transplantation¹³.

However, barcoding techniques carry intrinsic limitations. They are not yet fully quantitative, and in steady-state, the size of many clones is likely to be small and close to detection thresholds, so negative results must be interpreted with caution^{12,13}. This drawback is particularly relevant in the assessment of whether a stable haematopoietic clone is sustained HSC independently. The apex of a barcoded hierarchy would typically consist of very few cells and might therefore not be detected consistently. Similar to the current inability to reliably define HSCs and multipotent progenitors by virtue of cell-surface phenotypes alone, accurate determination of clone maintenance using barcoding remains a challenge. Another potential limitation of the retro-transposon system is the risk of insertional mutagenesis³³. Although this risk is likely to be small, no study to date has carefully assessed to what degree integration sites can influence clonal lineage readout.

Pei et al.¹⁴ generated a HSC-specific barcode to trace labelled output back to single HSCs. Consistent with previous studies^{27,28}, both myeloid-biased and lymphoid-biased HSC-derived clones were detected. However, lymphoid-biased HSC clones generally fail to serially repopulate²⁸ and do not sustain HSCs²⁵. An open and interesting question in unperturbed haematopoiesis is therefore to which extent barcoded clones with lymphoid bias maintain the originally labelled HSCs.

Overall, HSC barcoding reports similar HSC heterogeneity as observed in single-cell transplantation, indicating that the different HSC subsets identified through single-cell transplantation are actively employed in native haematopoiesis. This result is also consistent with the observation by Yu et al.¹⁵ that repopulation patterns of native HSCs identified by multicolour fluorescent tagging are preserved upon transplantation¹⁵. However, to fully reconcile the different findings, a systematic, integrated approach will probably be required, in which lineage tracing is combined with single-cell transplantation of labelled HSCs.

Studying haematopoiesis using lineage tracing

Single-cell-level analysis is essential to understand the heterogeneity that determines systems behaviour. In that regard, the collective properties of single cells should correspond to those of the entire system. To address this issue, lineage replenishment during steady-state haematopoiesis has been quantitatively measured through lineage tracing. This method permanently labels HSCs and their progeny using a Cre recombinase-inducible reporter specifically in HSCs³⁴. Essential for a successful experimental approach and correct data interpretation is that Cre is expressed in a representative manner in repopulating HSCs, covering their functional diversity. Likewise, it is important to avoid labelling of downstream progenitors. These caveats are highly relevant for recent studies that have sparked a debate as to whether unperturbed adult haematopoiesis depends on progenitor replenishment from HSCs^{12,13,32}. Using

Tie2-driven inducible Cre, Busch et al.³⁵ were able to label a subset (about 0.5%) of HSCs in young adult mice and observed that their replenishment of downstream progenitors and mature cells was very slow, suggesting, similarly to the barcoding studies of Sun et al.¹², that steady-state haematopoiesis is sustained largely independently of HSCs. In striking contrast, Sawai et al.³² observed near-complete labelling of all definitive haematopoietic lineages within a few months using *Pdzk1ip1*-driven Cre to quantitatively label HSCs, suggesting that steady-state blood lineage replenishment is highly HSC dependent. As described above, in the barcoding studies by Sun et al.¹², labelled HSCs may have gone undetected. In the work of Busch et al.³⁵, the limited *Tie2*-Cre labelling may have been restricted to a select subset of HSCs not including those actively contributing to steady-state haematopoiesis. This possibility is supported by the observation that few of the *Tie2*-Cre-labelled HSCs repopulated lymphoid lineages when transplanted at limiting dilution³⁵ compared to cells from an unselected HSC population²⁸. By contrast, Sawai et al.³² observed balanced repopulation across all definitive haematopoietic lineages, but in this case, the labelling and contribution of a subset of downstream progenitors cannot be excluded.

The importance of cautious and alternative interpretation of data generated by in vivo fate mapping is illustrated in Fig. 2. The same subset of haematopoietic lineages observed in a barcoding experiment (Fig. 2a) could be sustained by either a fate-restricted multipotent LT-HSC²⁵ (Fig. 2b), a fate-restricted multipotent progenitor with limited self-renewal (Fig. 2c) or an oligopotent progenitor with more limited self-renewal³¹ (Fig. 2d). Thus, a more definitive understanding of how native haematopoiesis is sustained will require an integrated approach that combines existing and potentially refined technologies, including in vivo fate mapping in conjunction with assays that determine lineage restriction and self-renewal properties of the cells sustaining clonal output.

Deciphering molecular and functional heterogeneity

Although the proposed haematopoiesis models have been valuable guides for genetic and pharmacological perturbation studies, they have also highlighted key technical limitations of the methodologies used in the field. In particular, progenitor cell populations may be functionally heterogeneous, even if identified through specific combinations of cell-surface markers^{8–10,20,36,37}. In addition, the in vivo and in vitro differentiation assays used to define the lineage potential of individual cells do not detect these properties with complete efficiency²⁶. Thus, even if the overall lineage potential of a progenitor population can be confidently measured in vitro, it is challenging to determine how many of these potentials reside in each individual progenitor cell, making assessment of progenitor population homogeneity challenging. Finally, although in vitro assays inform to some extent about lineage potential, they do not necessarily reflect the cellular fate in vivo.

The latter issue was addressed by in vivo molecular barcoding of CMPs³⁷ and LMPPs³⁶. These studies indicated that the vast majority of CMPs assumed either erythroid or myeloid fate in vivo, consistent with results obtained in vitro on CMPs subfractionated by cell sorting¹⁰. Similarly, most phenotypic LMPPs were observed to have a biased or restricted lineage fate in vivo, with the production of myeloid, dendritic and B cells highly correlated between cells derived from the same original LMPP³⁶. Although these studies have helped to unravel the functional heterogeneity within phenotypically defined progenitor populations, they also confirmed the existence of progenitors within the investigated CMP and LMPP populations adopting the predicted combined lineage fates.

Although similar progenitor populations have been identified in human haematopoiesis^{38–40}, differences may exist between mouse and human haematopoiesis. For example, recent studies have proposed that adult human lineage development occurs mainly through multipotent and unipotent progenitors, with oligopotent progenitor

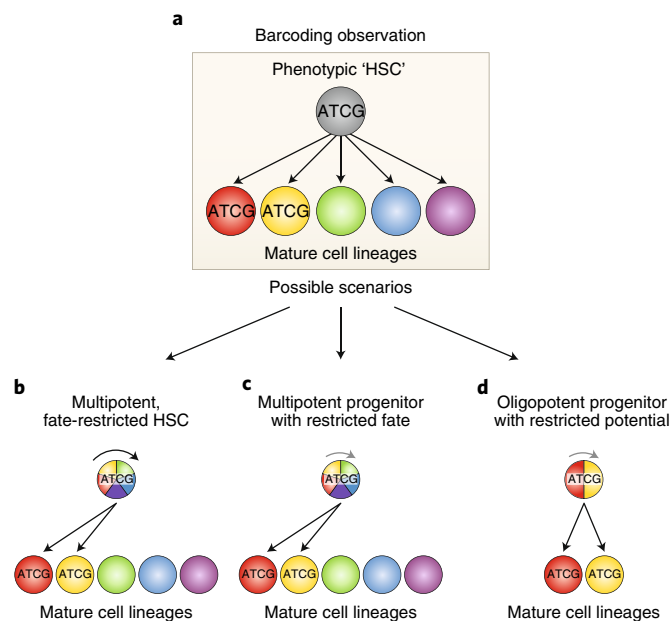


Fig. 2 | Possible scenarios underlying in vivo fate mapping outcomes.

a, An in vivo barcoding outcome in which clonal replenishment of a restricted set of lineages is observed during steady-state haematopoiesis, with the clone originating from a cell with an HSC immunophenotype. This outcome could be generated in several distinct scenarios. **b**, One option is that labelling occurred in a fate-restricted (that is, lineage-biased) multipotent repopulating HSC. In this scenario the cell of origin would be (1) multipotent when assayed in clonal lineage assays, and (2) possess extensive self-renewal potential (large black arrow) and be capable of long-term reconstitution in a serial transplantation assay. **c**, A second option is that the labelled clone originated from a multipotent progenitor with restricted self-renewal potential (small grey arrow) with little or no ability to long-term reconstitute upon transplantation, but revealing multipotency when assayed in in vitro clonal lineage assays. **d**, Finally, the cell of origin may represent a lineage-restricted progenitor, in vitro only possessing the same lineage potentials expressed in vivo, and with little or no long-term repopulating activity.

cells rarely being detected^{41,42}. These studies have the caveat that they are largely based on in vitro readouts that may not efficiently detect multiple different lineage potentials from the same single cell, emphasizing the need for further development of technologies to study human haematopoiesis, with ongoing and expanding gene therapy trials providing an opportunity to obtain information about human lineage commitment in vivo⁴³.

To access heterogeneity not resolved by surface markers, comprehensive profiling and clustering of both mouse and human haematopoietic progenitor cells using single-cell RNA sequencing (scRNAseq) has been performed^{44–47}. These studies readily detected cellular subsets with lineage-selective gene expression, including putative erythroid, platelet, monocytic and granulocytic lineage-restricted progenitors. However, they did not reproducibly identify the previously reported multipotent and oligopotent cell types, such as human LMPPs or MLPs and murine MPPs with distinct lineage potential⁴⁸. Diverging findings in different scRNAseq studies may in part reflect the effect of different library depths and analytical pipelines used. An alternative and highly sensitive approach to analyse transcriptional lineage programming at the single-cell level is targeted RT-PCR^{19,49}. Using this approach, co-expression of myeloid and lymphoid genes was consistently observed in single LMPPs, whereas Flt3-negative MPPs and HSCs co-expressed megakaryocyte/erythroid and myeloid programmes¹⁹, demonstrating distinct lineage programming

of MPP subsets, consistent with their in vivo lineage output⁴⁸. More recently, Gou et al.⁴⁹ used highly multiplexed single-cell quantitative reverse transcriptase (RT)-PCR to identify surface markers that separate transcriptionally and functionally distinct myeloid progenitor populations, demonstrating that targeted approaches to resolve cellular heterogeneity are an important and viable alternative to whole-transcriptome analysis. Single-cell RT-PCR and scRNAseq have also been applied in the study of ageing in haematopoiesis^{30,50,51}.

Transcriptional lineage priming has become a common tool for the assessment of lineage potential, based on the initial observation that single multipotent progenitors co-express genes and transcription factors characteristic of the lineages that they can differentiate into⁵². In particular, non-committed cellular states can be characterized by the co-expression of transcription factors critical for opposing lineage fates⁴⁵. Thus, the computational annotation of progenitor cells is based on the assumption that oligopotent and multipotent cell types will co-express the relevant lineage programmes and key regulators. However, it remains unclear to which extent the expression of lineage-associated genes accurately reflects cellular potential. Although the expression of lineage-specific genes in multipotent stem/progenitor cells can predict corresponding lineage potential, the opposite seems less clear^{19,20}. For example, even though HSCs are myeloid and megakaryocyte lineage primed, they do not display distinct priming for the lymphoid lineage potential that they possess^{18,19}. Upon differentiation into LMPPs, the acquisition of lymphoid lineage priming coincides with loss of the megakaryocyte-erythroid lineage potential^{18,19}, compatible with functional antagonism between these programmes contributing to concomitant increased megakaryocyte lineage priming and loss of lymphoid-biased HSCs with age³⁰. Accordingly, antagonistic lineage programmes may be sequentially, rather than simultaneously, expressed during differentiation, disguising latent lineage potential. Moreover, transcription levels stochastically vary within individual cells⁵³, requiring complex single-cell entropy analysis to reliably predict lineage potential⁵⁴. Assessment of chromatin states at the single-cell level may identify fate determinants that are not accessible at the transcriptome level and may provide information about the regulatory networks underlying cell-type specification^{55,56}. When using lineage priming as a proxy for lineage potential, these challenges and limitations have to be taken into account, and predictions based on molecular signatures should be functionally validated before interpreting their implications for lineage potential and cell fate.

The identification of myeloid progenitor subpopulations with basophil and eosinophil lineage priming^{44,45}, distinct from the neutrophil/monocyte priming that is generally used as a proxy for myeloid lineage potential, raised the issue of how these distinct myeloid lineages segregate. Computational modelling of human scRNAseq data suggested that all myeloid lineages derive from a single GMP, consistent with the predominant models of haematopoiesis⁴⁷. However, combined scRNAseq and single-cell quantitative RT-PCR analysis of murine myeloid progenitors revealed GMP heterogeneity based on *Gata1* expression, a finding used to prospectively separate *Gata1*-negative progenitors with neutrophil/monocyte-restricted potential from *Gata1*-positive eosinophil/mast cell-restricted potential¹⁷. Furthermore, eosinophil/mast cell potential and megakaryocyte/erythroid potential were observed to co-segregate from monocyte/neutrophil and lymphoid potential, indicating that *Gata1*-expressing myeloid lineages (eosinophils, basophils and mast cells) segregate from *Gata1*-negative myeloid lineages (neutrophils and monocytes) very early during lineage specification (Fig. 1c). These observations again underscore the importance of complementing computational analysis with the necessary functional validation.

Outlook and future perspectives

The application of classical and emerging single-cell-based technologies has substantially expanded and improved our knowledge

of the haematopoietic system. Single-cell transcriptome, epigenetic and transplantation analysis, as well as barcoding and in vitro clonal functional assays have enabled insights into the molecular heterogeneity that is difficult or impossible to gain using conventional population-based technologies. An important challenge will be to use these and other emerging single-cell technologies, such as mass cytometry, to unravel the functions of the haematopoietic system on a global scale. Importantly, as they all have intrinsic limitations, no technology to date is likely to fully resolve the complexity. Rather, effective integration of different single-cell molecular, genetic and functional methodologies will be required to understand the properties and roles played by distinct stem and progenitor populations in steady state, during tissue stress and cellular transformation.

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References

- Orkin, S. H. & Zon, L. I. Hematopoiesis: an evolving paradigm for stem cell biology. *Cell* **132**, 631–644 (2008).
- Laurenti, E. & Gottgens, B. From haematopoietic stem cells to complex differentiation landscapes. *Nature* **553**, 418–426 (2018).
- McCulloch, E. A. & Till, J. E. The radiation sensitivity of normal mouse bone marrow cells, determined by quantitative marrow transplantation into irradiated mice. *Radiat. Res.* **13**, 115–125 (1960).
- Spangrude, G. J., Heimfeld, S. & Weissman, I. L. Purification and characterization of mouse hematopoietic stem-cells. *Science* **241**, 58–62 (1988).
- Osawa, M., Hanada, K., Hamada, H. & Nakauchi, H. Long-term lymphohematopoietic reconstitution by a single CD34-low/negative hematopoietic stem cell. *Science* **273**, 242–245 (1996).
- Kiel, M. J. et al. SLAM family receptors distinguish hematopoietic stem and progenitor cells and reveal endothelial niches for stem cells. *Cell* **121**, 1109–1121 (2005).
- Metcalf, D. Hematopoietic cytokines. *Blood* **111**, 485–491 (2008).
- Akashi, K., Traver, D., Miyamoto, T. & Weissman, I. L. A clonogenic common myeloid progenitor that gives rise to all myeloid lineages. *Nature* **404**, 193–197 (2000).
- Kondo, M., Weissman, I. L. & Akashi, K. Identification of clonogenic common lymphoid progenitors in mouse bone marrow. *Cell* **91**, 661–672 (1997).
- Pronk, C. J. et al. Elucidation of the phenotypic, functional, and molecular topography of a myeloerythroid progenitor cell hierarchy. *Cell Stem Cell* **1**, 428–442 (2007).
- Lu, R., Neff, N. F., Quake, S. R. & Weissman, I. L. Tracking single hematopoietic stem cells in vivo using high-throughput sequencing in conjunction with viral genetic barcoding. *Nat. Biotechnol.* **29**, 928–933 (2011).
- Sun, J. et al. Clonal dynamics of native haematopoiesis. *Nature* **514**, 322–327 (2014).
- Rodriguez-Fraticelli, A. E. et al. Clonal analysis of lineage fate in native haematopoiesis. *Nature* **553**, 212–216 (2018).
- Pei, W. et al. Polylox barcoding reveals haematopoietic stem cell fates realized in vivo. *Nature* **548**, 456–460 (2017).
- Yu, V. W. et al. Epigenetic memory underlies cell-autonomous heterogeneous behavior of hematopoietic stem cells. *Cell* **168**, 944–945 (2017).
- Jones, R. J. & Armstrong, S. A. Cancer stem cells in hematopoietic malignancies. *Biol. Blood Marrow Transplant.* **14**, 12–16 (2008).
- Drissen, R. et al. Distinct myeloid progenitor-differentiation pathways identified through single-cell RNA sequencing. *Nat. Immunol.* **17**, 666–676 (2016).
- Adolfsson, J. et al. Identification of Flt3⁺ lympho-myeloid stem cells lacking erythro-megakaryocytic potential: a revised road map for adult blood lineage commitment. *Cell* **121**, 295–306 (2005).
- Mansson, R. et al. Molecular evidence for hierarchical transcriptional lineage priming in fetal and adult stem cells and multipotent progenitors. *Immunity* **26**, 407–419 (2007).
- Arinobu, Y. et al. Reciprocal activation of GATA-1 and PU.1 marks initial specification of hematopoietic stem cells into myeloerythroid and myelolymphoid lineages. *Cell Stem Cell* **1**, 416–427 (2007).
- Miyawaki, K. et al. CD41 marks the initial myelo-erythroid lineage specification in adult mouse hematopoiesis: redefinition of murine common myeloid progenitor. *Stem Cells* **33**, 976–987 (2015).
- Yanez, A. et al. Granulocyte-monocyte progenitors and monocyte-dendritic cell progenitors independently produce functionally distinct monocytes. *Immunity* **47**, 890–902.e4 (2017).
- Wilson, A. et al. Hematopoietic stem cells reversibly switch from dormancy to self-renewal during homeostasis and repair. *Cell* **135**, 1118–1129 (2008).
- Bernitz, J. M., Kim, H. S., MacArthur, B., Sieburg, H. & Moore, K. Hematopoietic stem cells count and remember self-renewal divisions. *Cell* **167**, 1296–1309.e10 (2016).
- Carrelha, J. et al. Hierarchically related lineage-restricted fates of multipotent haematopoietic stem cells. *Nature* **554**, 106–111 (2018).
- Rieger, M. A., Hoppe, P. S., Smejkal, B. M., Eitelhuber, A. C. & Schroeder, T. Hematopoietic cytokines can instruct lineage choice. *Science* **325**, 217–218 (2009).
- Muller-Sieburg, C. E., Cho, R. H., Thoman, M., Adkins, B. & Sieburg, H. B. Deterministic regulation of hematopoietic stem cell self-renewal and differentiation. *Blood* **100**, 1302–1309 (2002).
- Dykstra, B. et al. Long-term propagation of distinct hematopoietic differentiation programs in vivo. *Cell Stem Cell* **1**, 218–229 (2007).
- Sanjuan-Pla, A. et al. Platelet-biased stem cells reside at the apex of the haematopoietic stem-cell hierarchy. *Nature* **502**, 232–236 (2013).
- Grover, A. et al. Single-cell RNA sequencing reveals molecular and functional platelet bias of aged haematopoietic stem cells. *Nat. Commun.* **7**, 11075 (2016).
- Yamamoto, R. et al. Clonal analysis unveils self-renewing lineage-restricted progenitors generated directly from hematopoietic stem cells. *Cell* **154**, 1112–1126 (2013).
- Sawai, C. M. et al. Hematopoietic stem cells are the major source of multilineage hematopoiesis in adult animals. *Immunity* **45**, 597–609 (2016).
- Copeland, N. G. & Jenkins, N. A. Harnessing transposons for cancer gene discovery. *Nat. Rev. Cancer* **10**, 696–706 (2010).
- Kretzschmar, K. & Watt, F. M. Lineage tracing. *Cell* **148**, 33–45 (2012).
- Busch, K. et al. Fundamental properties of unperturbed haematopoiesis from stem cells in vivo. *Nature* **518**, 542–546 (2015).
- Naik, S. H. et al. Diverse and heritable lineage imprinting of early hematopoietic progenitors. *Nature* **496**, 229–232 (2013).
- Perie, L., Duffy, K. R., Kok, L., de Boer, R. J. & Schumacher, T. N. The branching point in erythro-myeloid differentiation. *Cell* **163**, 1655–1662 (2015).
- Manz, M. G., Miyamoto, T., Akashi, K. & Weissman, I. L. Prospective isolation of human clonogenic common myeloid progenitors. *Proc. Natl Acad. Sci. USA* **99**, 11872–11877 (2002).
- Doulatov, S. et al. Revised map of the human progenitor hierarchy shows the origin of macrophages and dendritic cells in early lymphoid development. *Nat. Immunol.* **11**, 585–593 (2010).
- Goardon, N. et al. Coexistence of LMPP-like and GMP-like leukemia stem cells in acute myeloid leukemia. *Cancer Cell* **19**, 138–152 (2011).
- Notta, F. et al. Distinct routes of lineage development reshape the human blood hierarchy across ontogeny. *Science* **351**, aab2116 (2016).
- Karamitros, D. et al. Single-cell analysis reveals the continuum of human lympho-myeloid progenitor cells. *Nat. Immunol.* **19**, 85–97 (2018).
- Biasco, L. et al. In vivo tracking of human hematopoiesis reveals patterns of clonal dynamics during early and steady-state reconstitution phases. *Cell Stem Cell* **19**, 107–119 (2016).
- Paul, F. et al. Transcriptional heterogeneity and lineage commitment in myeloid progenitors. *Cell* **164**, 325 (2016).
- Olsson, A. et al. Single-cell analysis of mixed-lineage states leading to a binary cell fate choice. *Nature* **537**, 698–702 (2016).
- Nestorowa, S. et al. A single-cell resolution map of mouse hematopoietic stem and progenitor cell differentiation. *Blood* **128**, e20–e31 (2016).
- Velten, L. et al. Human haematopoietic stem cell lineage commitment is a continuous process. *Nat. Cell Biol.* **19**, 271–281 (2017).
- Pietras, E. M. et al. Functionally distinct subsets of lineage-biased multipotent progenitors control blood production in normal and regenerative conditions. *Cell Stem Cell* **17**, 35–46 (2015).
- Guo, G. et al. Mapping cellular hierarchy by single-cell analysis of the cell surface repertoire. *Cell Stem Cell* **13**, 492–505 (2013).
- Warren, L. A. et al. Transcriptional instability is not a universal attribute of aging. *Aging Cell* **6**, 775–782 (2007).
- Kowalczyk, M. S. et al. Single-cell RNA-seq reveals changes in cell cycle and differentiation programs upon aging of hematopoietic stem cells. *Genome Res.* **25**, 1860–1872 (2015).
- Hu, M. et al. Multilineage gene expression precedes commitment in the hemopoietic system. *Genes Dev.* **11**, 774–785 (1997).
- Raj, A. & van Oudenaarden, A. Nature, nurture, or chance: stochastic gene expression and its consequences. *Cell* **135**, 216–226 (2008).
- Teschendorff, A. E. & Enver, T. Single-cell entropy for accurate estimation of differentiation potency from a cell's transcriptome. *Nat. Commun.* **8**, 15599 (2017).
- Corces, M. R. et al. Lineage-specific and single-cell chromatin accessibility charts human hematopoiesis and leukemia evolution. *Nat. Genet.* **48**, 1193–1203 (2016).
- Buenrostro, J. D. et al. Integrated single-cell analysis maps the continuous regulatory landscape of human hematopoietic differentiation. *Cell* **173**, 1535–1548.e16 (2018).

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Competing interests

The authors declare no competing interests.

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